Bernoulli extremizers for the independent Berry–Esseen ratio

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Abstract

We study the extremal structure of the Berry–Esseen ratio for independent centered two-point (Bernoulli) random variables, allowing deterministic zero summands. First, for every $n \geq 1$, we prove that every array maximizing the Berry–Esseen ratio among n-summand two-point arrays with a common support diameter consists of $m \leq n$ identically distributed nondegenerate two-point summands and $n - m$ zero summands. Second, for at most two two-point summands with arbitrary support diameters, we prove that the maximum ratio equals the one-summand constant $C_1 \approx 0.3703515235$, attained only when one summand is zero. Third, we determine the asymptotic behavior of the optimal active count m_N in the binomial envelope $P_N = \max_{1 \leq m \leq N} B_m$: a uniform seventh-order Edgeworth–Euler–Maclaurin expansion and a quadratic-field separation argument in $\mathbb{Q}(\sqrt{10})$ prove that for all sufficiently large N, m_N is unique, the number of zero summands $N - m_N$ belongs to $\{0, 1, 2, 3, 4\}$ with explicit limiting frequencies, and successive record active sizes have gaps in $\{2, 3, 5\}$. Fourth, for two-point arrays with large count variance $A = \sum_{i=1}^{n} p_i(1 - p_i)$ and small diameter deviations $e_i = g_i/h_0 - 1$, we prove that replacing all diameters by their variance-weighted mean h_0 strictly increases the ratio by at least $\kappa\sigma^2/(\sigma + \eta)$, where $\sigma^2 = \sum_{i=1}^{n} p_i(1 - p_i)e_i^2$ and $\eta = \max_i |e_i|$, and we construct an explicit family showing that the rate $\sigma^2/(\sigma + \eta)$ is optimal in the absence of a lower bound on the individual probabilities.

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1 Introduction and main results

1.1 Formulation of the extremal problem

Let $X_1, \dots, X_n$ be independent centered real random variables with positive total variance and finite third absolute moments, and set

$$V = \sum_{i=1}^{n} \mathbb{E}X_i^2 > 0, \qquad R = \sum_{i=1}^{n} \mathbb{E}|X_i|^3 < \infty,$$
$$\mathcal{C}(X_1, \dots, X_n) = \frac{V^{3/2}}{R} \sup_{y \in \mathbb{R}} \left| \mathbb{P}\left\{ \sum_{i=1}^{n} X_i \le y \right\} - \Phi\Big(\frac{y}{\sqrt{V}} \Big) \right|,$$

where Φ and ϕ are the standard normal distribution function and density. A *two-point* (or *centered Bernoulli*) summand has the form $X_i = h_i(B_i - p_i)$, where $h_i > 0$ is its support diameter and $B_i \sim \mathrm{Bern}(p_i)$ $(0 \le p_i \le 1)$; the boundary values $p_i \in \{0, 1\}$ yield a deterministic zero summand. For two-point summands,

$$V = \sum_{i=1}^{n} h_i^2 p_i(1-p_i), \qquad R = \sum_{i=1}^{n} h_i^3 p_i(1-p_i)\{p_i^2 + (1-p_i)^2\}. \tag{1.1}$$

By Esseen [4, pp. 160–163], n independent copies of a two-point summand with upper atom probability $p_{\mathrm{E}} = (4 - \sqrt{10})/2$ achieve $\lim_{n\to\infty} \mathcal{C} = C_{\mathrm{E}} := (3 + \sqrt{10})/(6\sqrt{2\pi}) \approx 0.409732$. For finite $m \ge 1$, Schulz [7, Theorem 1] proved that the m-summand i.i.d. binomial maximum $B_m := \max_{0<p<1} \mathcal{C}(B_1 - p, \dots, B_m - p)$ satisfies $B_m < C_{\mathrm{E}}$, and a companion paper [1, Theorem 1.2(ii)] extended the upper bound $\mathcal{C} \le C_{\mathrm{E}}$ to all finite common-diameter two-point arrays.

Whereas Bell [1] bounds the ratio $\mathcal{C}$ from above by C_{E}, the present paper characterizes the *maximizers* of $\mathcal{C}$ at a fixed number of summands n, analyzes how $\mathcal{C}$ behaves under perturbations of the probabilities and support diameters, and explains why setting a bounded number of coordinates to zero can strictly increase the finite-n binomial maximum.

1.2 Main results

Let B_m denote the maximal ratio for m identically distributed active two-point summands, and let C_n^{com} denote the maximum of $\mathcal{C}(X_1, \dots, X_n)$ over arrays of at most n two-point summands sharing a common support diameter (permitting zero summands). For $n = 1$,

$$B_1 = C_1 = \max_{0<p<1} \frac{\sqrt{p(1-p)}}{p^2 + (1-p)^2} \left\{ \Phi\left(\sqrt{\frac{p}{1-p}} \right) - p \right\} = 0.370351523474378920\ldots, \tag{1.2}$$

which is also the unrestricted one-summand Berry–Esseen constant [2, Theorem 1].

Our first theorem classifies all common-diameter maximizers at every finite $n \geq 1$.

Theorem 1.1 (Classification of common-diameter maximizers)**.** *For every integer $n \geq 1$,*

$$C_n^{\mathrm{com}} = \max_{1 \leq m \leq n} B_m.$$

Moreover, every n-summand common-diameter array attaining C_n^{com} consists of $m \in \{1, \dots, n\}$ identically distributed nondegenerate two-point summands and $n - m$ zero summands.

When $n = 2$, we remove the common-diameter hypothesis completely and determine the exact maximum over all pairs of two-point summands with arbitrary diameters.

Theorem 1.2 (Two two-point summands with arbitrary diameters)**.** *Over arrays of at most two independent two-point summands with arbitrary support diameters, the maximum of $\mathcal{C}(X_1, X_2)$ is C_1, and every attaining array has exactly one nonconstant summand. If both summands are required to have positive variance, the supremum remains C_1 and is not attained.*

Our third theorem quantifies the strict improvement obtained when a two-point array with large total count variance $A = \sum_{i=1}^n p_i(1 - p_i)$ and small diameter spread is replaced by the common-diameter array with the same probabilities $p_1, \dots, p_n$.

Theorem 1.3 (Strict improvement under common-diameter replacement)**.** *Fix $c_0 > 0$. There exist constants $A_0, \sigma_0, \eta_0, \kappa > 0$, depending only on c_0, with the following property. Let $X_i = g_i(B_i - p_i)$ $(1 \leq i \leq n)$ be independent with $g_i > 0$ and $0 < p_i < 1$, and set*

$$a_i = p_i(1 - p_i), \qquad A = \sum_{i=1}^n a_i, \qquad h_0 = \frac{\sum_{i=1}^n a_i g_i}{A},$$
$$e_i = \frac{g_i}{h_0} - 1, \qquad \sigma^2 = \sum_{i=1}^n a_i e_i^2, \qquad \eta = \max_{1 \leq i \leq n} |e_i|.$$

If $A \geq A_0$, $0 < \sigma \leq \sigma_0$, $\eta \leq \eta_0$, and $\mathcal{C}(X_1, \dots, X_n) \geq c_0$, then

$$\mathcal{C}(B_1 - p_1, \dots, B_n - p_n) - \mathcal{C}(X_1, \dots, X_n) \geq \kappa \frac{\sigma^2}{\sigma + \eta} > 0.$$

If in addition all probabilities p_i lie in a fixed compact subset $J \subset (0, 1)$, then the improvement is at least $\kappa_J \sigma$ for a constant $\kappa_J > 0$ depending only on c_0 and J. Furthermore, the exponent $\sigma^2/(\sigma + \eta)$ cannot be improved to σ when rare probabilities $p_i \to 0$ are permitted.

Finally, in Section 4 (Theorem 4.1) we determine the asymptotic behavior of the padded binomial maximum $P_N = \max_{1 \leq m \leq N} B_m$: for all sufficiently large N, P_N has a unique maximizing active count m_N, the number of zero summands $j_N = N - m_N$ belongs to $\{0, 1, 2, 3, 4\}$ with explicit limiting frequencies, and the gaps between successive record active counts belong to $\{2, 3, 5\}$.

1.3 Open questions and overview of the methods

The results of this paper motivate two open structural conjectures for fixed $n \geq 1$:

(1) *Global optimality of common diameters among two-point arrays:* Theorem 1.2 establishes that unequal diameters cannot beat a common diameter for $n = 2$, and Theorem 1.3 establishes this locally for large count variance A and small diameter spread. However, as shown in the companion numerical note, unequal diameters with rational ratios such as $(1, 1, 1, 2)$ at $n = 4$ can create multi-summand atom collisions that are strict local maxima of $\mathcal{C}$ (though with ratio $\approx 0.2825 < P_4 \approx 0.4009$). Proving globally that $\sup \mathcal{C}(X_1, \ldots, X_n) = P_n$ across all n-summand two-point arrays with arbitrary diameters for $n \geq 3$ remains open.

(2) *Reduction from general summands to two-point summands:* Combining Theorem 1.1 with the small-Lyapunov phase-collapse and lattice-concavity theorems of the companion structure paper proves that every unrestricted n-summand global maximizer with Lyapunov ratio below an absolute threshold $\varepsilon_0 > 0$ is a padded i.i.d. two-point array. Extending this classification to all $n \geq 1$ without a small-Lyapunov restriction would imply $C_{\text{ind}} = C_{\text{E}}$.

For every finite-support sum S, we use the one-sided branch decomposition

$$\sup_{y\in\mathbb{R}} \left| F(y) - \Phi\left(\frac{y}{\sqrt{V}}\right) \right| = \sup_{y\in\mathbb{R}} \max \left\{ F(y) - \Phi\left(\frac{y}{\sqrt{V}}\right), \Phi\left(\frac{y}{\sqrt{V}}\right) - F^{-}(y) \right\}, \tag{1.3}$$

where $F(y) = \mathbb{P}(S \leq y)$ and $F^{-}(y) = \mathbb{P}(S < y)$. Reflection $X_i \mapsto -X_i$ interchanges the two branches, and between consecutive atoms the closed discrepancy $F(y) - \Phi(y/\sqrt{V})$ is strictly decreasing, so its maximum is attained at a support atom.

The paper is organized as follows:

- **Section 2 (Proof of Theorem 1.1):** Fixing the mean μ, variance V, and closed integer threshold k of a common-diameter maximizer, we maximize $\sum_i p_i^3$ on the connected component K of the zero-deficit set $Z = \{p : H_C(p) = 0\}$. Varying three probabilities along their elementary-symmetric (e_1, e_2) fiber and using a uniform third-difference bound (Lemma 2.1) reduces the cubic maximizer on K to $\text{Bin}(M, p)+\text{Bern}(x)$, where a certified curvature inequality (Lemma 2.2) rules out $p \neq x$ and proves that the i.i.d. point is isolated in Z, forcing K to be a singleton.

- **Section 3 (Proof of Theorem 1.2):** For two two-point summands with diameters 1 and $t \in (0, 1)$, variational contact bounds confine any stationary point with $\mathcal{C} \geq 37/100$ to a compact variance domain $V > 1369/3319684$, where a 37,287-leaf certified interval partition rules out all three interior threshold branches.

- **Section 4 (Binomial phase and zero padding):** We derive a uniform seventh-order Edgeworth–Euler–Maclaurin expansion for $D_{m,k}(v) = F_{m,v}(k) - \Phi(-cd)$, compute the $O(m^{-2})$ and $O(m^{-3})$ phase expansions of B_m, determine the five-parabola lower envelope on $[-1/2, 1/2]$, and use a field-norm lower bound in $\mathbb{Q}(\sqrt{10})$ to control the $O(N^{-1})$ boundary drift.

- **Section 5 (Proof of Theorem 1.3):** Using uniform Poisson-binomial local and jittered Edgeworth expansions with no individual probability floor, we bound the unconditioned weighted perturbation moments $N_j(k) = \mathbb{E}[T^j \mathbf{1}_{\{K=k\}}]$, localize any threshold with $\mathcal{C} \geq c_0$ to the central bulk, establish the conditional L^1 lower bound $\mathbb{E}(|T| \mid K = k) \geq c\ell/\sqrt{A}$ by a perfect-matching coupling of two rejective-sampling copies, and construct the rare-coordinate example proving optimality of the rate $\sigma^2/(\sigma + \eta)$.

2 Classification of common-diameter maximizers

Scale the common diameter to 1 and write

$$\mu=\sum_{i=1}^{n}p_i,\qquad V=\sum_{i=1}^{n}p_i(1-p_i),\qquad R=\sum_{i=1}^{n}\{p_i-3p_i^2+4p_i^3-2p_i^4\},\qquad F_k=\mathbb{P}\left\{\sum_{i=1}^{n}B_i\le k\right\}.$$

Since $R\ge V/2$, the ratio satisfies $\mathcal{C}\le 2\sqrt{V}$, so its extension by zero at $V=0$ is continuous on $[0,1]^n$ and $C=C_n^{\mathrm{com}}$ is attained. At any selected closed integer threshold k, define the deficit function

$$H_C(p)=\frac{CR}{V^{3/2}}+\Phi\left(\frac{k-\mu}{\sqrt{V}}\right)-F_k. \tag{2.1}$$

By definition of $C=C_n^{\mathrm{com}}$, $H_C(p)\ge 0$ at every $p\in[0,1]^n$ with $V>0$, and $H_C(p)=0$ at every maximizing array at its attaining threshold k.

Our proof combines three-coordinate elementary-symmetric probability fibers (compare Hoeffding [5, eq. (10)] and Mattner [6, Theorems 1.1–1.2 and Lemma 2.1]) with a connected-component isolation argument showing that every maximizer—not merely an extreme tie-broken maximizer—consists of identically distributed active coordinates and zeros.

2.1 Uniform third-difference bound for Poisson-binomial laws

For the point mass function $c_j=\mathbb{P}(\sum_i B_i=j)$ $(j\in\mathbb{Z})$ of a finite Poisson-binomial sum, put $\Delta^3 c_j=c_j-3c_{j-1}+3c_{j-2}-c_{j-3}$.

Lemma 2.1. *If a finite Poisson-binomial sum has variance W, then*

$$\sup_{j\in\mathbb{Z}}|\Delta^3 c_j|<\frac{16/5}{(1+W)^{3/2}}.$$

Proof. For $0\le W\le 1/4$, take the more probable outcome of each coordinate as a base state, let J count departures from that base state, and set $\epsilon_i=\min(p_i,1-p_i)$, $E=\sum_i\epsilon_i\le 2W\le 1/2$, and $Q=\sum_i\epsilon_i/(1-\epsilon_i)\in[W,1]$. Then $\mathbb{P}(J=0)\ge\max(1-E,e^{-Q})\ge 1/2$ and $\mathbb{P}(J=1)=Q\mathbb{P}(J=0)\ge We^{-W}$. At the two indices j where the base state contributes ±3 to $\Delta^3 c_j$, every $J=1$ configuration contributes with opposite sign and magnitude at least 1, giving $|\Delta^3 c_j|\le 3-4We^{-W}\le 3-4W+4W^2$; at all other indices, $|\Delta^3 c_j|\le 1+4W\le 2<(16/5)(5/4)^{-3/2}$. Using $(1+W)^{-3/2}\ge h(W):=1-3W/2+15W^2/8-35W^3/16$, the polynomial $(16/5)h(W)-(3-4W+4W^2)$ decreases on $[0,1/4]$ to $1/64>0$ at $W=1/4$.

For $W\ge 1/4$, Fourier inversion gives

$$|\Delta^3 c_j|\le I(W):=\frac{16}{\pi}\int_0^{\pi/2}\sin^3 u\,e^{-2W\sin^2 u}\,du.$$

Writing $J_r=\int_0^{\pi/2}\sin^r u\,e^{-2W\sin^2 u}\,du$, integration by parts gives $(5+4W)J_5=4J_3+4WJ_7$, so

$$\frac{d}{dW}\log\{I(W)(1+W)^{3/2}\}\le-\frac{8}{5+4W}+\frac{3}{2(1+W)}<0,$$

and at $W=1/4$ the degree-six Taylor polynomial of $e^{-(\sin^2 u)/2}$ gives

$$I(1/4)(5/4)^{3/2}\le\frac{10\sqrt{5}}{\pi}\,\frac{911264}{2027025}<\frac{16}{5},$$

using $\pi>6283/2000$. □

2.2 Reduction on a connected component of the zero set

Fix a maximizing array $p^{(0)} \in [0,1]^n$ and its mean μ, variance V, and attaining closed threshold k. Consider the compact zero set on the (μ, V) fiber:

$$Z = \left\{ p \in [0,1]^n : \sum_{i=1}^n p_i = \mu,\ \sum_{i=1}^n p_i(1-p_i) = V,\ H_C(p) = 0 \right\}.$$

Let $K \subset Z$ be the connected component of Z containing $p^{(0)}$, and choose a point $p^* \in K$ maximizing $\sum_{i=1}^n p_i^3$ on K. For any three coordinates, let e_1, e_2, e_3 be their elementary symmetric polynomials. Holding e_1, e_2 fixed preserves both their sum e_1 and their variance $e_1 - e_1^2 + 2e_2$, while their contribution to R is

$$R_3 = e_1 - 3e_1^2 + 4e_1^3 - 2e_1^4 + (6 - 12e_1 + 8e_1^2)e_2 - 4e_2^2 + 4(3 - 2e_1)e_3.$$

Thus both F_k and R—and hence H_C—are affine in e_3 along the (e_1, e_2) fiber. If three coordinates of p^* were distinct and in $(0,1)$, e_3 could move in both directions along the fiber; a nonzero slope of H_C would violate $H_C \geq 0$, while a zero slope would yield a continuous segment in Z (and hence in the connected component K) along which $p_1^3 + p_2^3 + p_3^3 = e_1^3 - 3e_1e_2 + 3e_3$ strictly increases. Consequently, p^* has at most two distinct interior probability values $0 < a < b < 1$.

Suppose both a and b occurred with multiplicity at least two in p^*. Let d_j be the point mass function of the $n-3$ coordinates outside a selected triple; the slope of H_C with respect to e_3 is $L = 4C(3-2e_1)/V^{3/2} + \Delta^2 d_k$. Since (a,a,b) and (a,b,b) lie at opposite endpoints of their three-root (e_1, e_2) fibers, $H_C \geq 0$ forces $L_{aab} \leq 0 \leq L_{abb}$. Deleting two copies of a and two copies of b, and letting c_j be the point mass function of the remaining $n-4$ coordinates (with variance $W \leq V-1$), subtraction gives

$$0 \leq L_{abb} - L_{aab} = (b-a)\left\{ \Delta^3 c_k - \frac{8C}{V^{3/2}} \right\}.$$

For $n \geq 5$, $C \geq 2/5$ (as verified by the explicit 5-coordinate example below), so $8C/V^{3/2} \geq (16/5)(1+W)^{-3/2}$, contradicting Lemma 2.1. For $n = 4$, the only $(2,2)$ pattern has deterministic deleted sum ($W = 0$, $|\Delta^3 c_k| \leq 3$), whereas $8C/V^{3/2} \geq 8C > 3$ since $C \geq 199/500$. Hence at least one of the two interior values a, b has multiplicity one.

2.3 Strict curvature of the one-exception fiber

The remaining unequal interior configuration has the form $\mathrm{Bin}(M,p) + \mathrm{Bern}(x)$ with $p \neq x$. Put $q = 1-p$, $m = M-1$, $d_j = \mathbb{P}(\mathrm{Bin}(m,p) = j)$, $W = mpq$, and

$$V = x(1-x) + Mpq, \qquad z = \frac{k - x - Mp}{\sqrt{V}}, \qquad \xi = d_{k-1} - d_k,$$
$$S_2 = (1-2x)^2 + (1-2x)(1-2p) + (1-2p)^2.$$

Varying the exceptional coordinate x and one copy of p as $\bar{p} + d, \bar{p} - d$ with $u = d^2$ gives $F_k' = \xi$, $V' = -2$, $R' = -2S_2$, and $R'' = -8$. Eliminating R at a stationary point $H_C'(u) = 0$ yields

$$V^2 H_C'' = 5V\xi - C\left(8\sqrt{V} + \frac{2S_2}{\sqrt{V}}\right) - z\phi(z)(2+z^2). \tag{2.2}$$

Lemma 2.2 (Certified curvature inequality)**.** *Set* $c_* = 199/500$. *For all* $0 < p, x < 1$, $pq \le W \le 3$, $0 \le f \le 1$, *and* $A_0 = W - (1-f)q^2 \ge 0$, *define*

$$B_0 = p^2 + (1-f)(W+2pq), \qquad V = x(1-x) + W + pq, \qquad z = \frac{f(W+pq)/(1-qf) - x}{\sqrt{V}}.$$

Then

$$\frac{2c_*}{\sqrt{V}}\{4W + 2 + (1-2x)(1-2p)\} + z\phi(z)(2+z^2) - \frac{5VfB_0}{(2-f)B_0 + A_0} > 0. \tag{2.3}$$

Furthermore, every two-summand unit-diameter array has ratio strictly less than c_*.

Lemma 2.2 is verified on $V \le 1/5$ and $W \ge 3$ by the elementary bounds

$$\frac{8c_*}{\sqrt{3}} - \frac{4}{e\sqrt{\pi}} - 1 > \frac{1}{247}, \qquad 8c_*\sqrt{3} - \frac{10}{\pi} - \frac{4}{e\sqrt{\pi}} > 1, \qquad 8c_*\sqrt{7/2} - \frac{35}{3\pi} - \frac{4}{e\sqrt{\pi}} > \frac{44}{45}, \tag{2.4}$$

and on $V \ge 1/5$, $pq \le W \le 3$ by a 192-bit Arb interval cover (550,563 processed boxes, 275,283 accepted boxes for (2.3), and 13,641 processed boxes, 6,822 accepted boxes for the two-summand bound; see `certificates/structure/README.md`). Consequently, for every $C \ge c_*$, (2.2) is strictly negative for $M \ge 2$, while the two-summand bound excludes $M = 1$.

2.4 Isolation of the i.i.d. point and completion of Theorem 1.1

By the preceding reductions, for $n \ge 3$ with $C \ge c_*$, the cubic maximizer p^* on the connected component $K \subset Z$ consists of $m \ge 3$ active coordinates all equal to $p \in (0,1)$ and $n-m$ deterministic coordinates in $\{0,1\}$. We now show that p^* is an isolated point of Z.

Along the two-coordinate exchange path $p \pm d$ ($u = d^2$) within the active group, $H_C \ge 0$ implies $\Lambda := \partial_u H_C(0) \ge 0$. If $\Lambda = 0$, then (2.2) with $M = m - 1 \ge 2$ gives $\partial_u^2 H_C(0) < 0$, violating $H_C \ge 0$ for small $u > 0$. Thus $\Lambda > 0$, and by permutation symmetry of the m active coordinates, the quadratic form of H_C in mean-zero active perturbations y_{cen} is $(\Lambda/2)\|y_{\mathrm{cen}}\|^2$.

For any nearby point in Z obtained by perturbations $a_i \ge 0$ of the zero coordinates, $-b_j \le 0$ of the unit coordinates, and y_l of the m active coordinates, put $a = \sum a_i$, $b = \sum b_j$, and $t = a + b$. The fixed mean and variance constraints give

$$\sum_{l=1}^{m} y_l = b - a, \qquad \sum_{l=1}^{m} y_l^2 = 2pa + 2(1-p)b - \sum_i a_i^2 - \sum_j b_j^2,$$

so $\|y_{\mathrm{cen}}\|^2 \ge c_p t - O(t^2)$ with $c_p > 0$. Since the first-order derivatives of H_C with respect to activating boundary coordinates (with uniform active compensation) are nonnegative, Taylor expansion of H_C on the (μ, V) fiber yields $H_C \ge (\Lambda/2)c_p t - O(t^{3/2}) > 0$ for all small $t > 0$, while $t = 0$ forces $y = 0$. Thus p^* is isolated in Z, so its connected component K is the singleton $\{p^*\}$, and the original maximizer $p^{(0)}$ equals p^*.

To supply the lower bounds $C_n^{\mathrm{com}} > c_* = 199/500$ for $n \ge 3$ and $C_n^{\mathrm{com}} > 2/5$ for $n \ge 5$, we evaluate two explicit rational binomial examples:

Active count m	p	Closed threshold k	Certified lower bound on $\mathcal{C}$
3	43/100	1	$0.3981444741 > 199/500$
5	9/20	2	$0.4009890598 > 2/5$.

2.5 The two-coordinate case ($n = 2$)

Finally, for $n = 2$ with probabilities (p, x), set $s = p + x$ and $w = px$, so that $V = s - s^2 + 2w$ and $R = s - 3s^2 + 4s^3 - 2s^4 + (6 - 12s + 8s^2)w - 4w^2$. At threshold $k \in \{0, 1, 2\}$, let $z = (k - s)/\sqrt{V}$, $D = F_k - \Phi(z)$, and $C = DV^{3/2}/R$. Off the diagonal $p = x$, $(p, x) \mapsto (s, w)$ is a diffeomorphism, and stationarity of C gives the numerators

$$G_w = R(VF_w + z\phi(z) + 3D) - DVR_w = 0,$$
$$G_s = 2VRF_s + 2\sqrt{V}R\phi(z) + Rz\phi(z)V_s + 3DRV_s - 2DVR_s = 0,$$

while at $G_w = 0$ the second derivative C_{ww} has the sign of $N = R\{6F_wV + z\phi(z)(3 + z^2) + 3D\} + 8DV^2$. The interval certificate `certificates/structure/two_equal_gaps.json` covers the square $[0, 1]^2$ (4,683 off-diagonal boxes and 195 diagonal boxes), certifying that at every point either $C < 37/100$, $0 \notin G_w$, $0 \notin G_s$, or $N > 0$. Since the one-coordinate example $p = 2/5$ achieves $\mathcal{C} > 37/100$, every $n = 2$ maximizer has exactly one active coordinate. This completes the proof of Theorem 1.1.

3 Two two-point summands with arbitrary diameters

We now prove Theorem 1.2. Normalizing $V = 1$, any maximizing sequence of two-point pairs with $\mathcal{C} \geq c > 0$ has $1/\sqrt{2} \leq R \leq 1/c$, uniformly integrable second moments $\sum_i \mathbb{E}[X_i^2 \mathbf{1}_{\{|X_i| > L\}}] \leq 1/(cL)$, and bounded thresholds y. Coordinatewise weak limits of two-point laws remain supported on at most two points, preserve $V = 1$, and attain the two-summand binary supremum $C_{2,\mathrm{bin}}$.

3.1 Top threshold and equal-diameter boundary

For a centered two-point variable on $\{-b, a\}$ $(a, b > 0)$ with variance $v = ab$, its third absolute moment satisfies $\rho = ab(a^2 + b^2)/(a + b) \geq 2(\sqrt{2} - 1)v^2/a$. For two such summands with upper support endpoint $L = a_1 + a_2$, Cauchy–Schwarz gives $R \geq 2(\sqrt{2} - 1)V^2/L$. At the top closed threshold $y = L$, where $F(L) = 1$, the ratio is therefore bounded above by

$$\frac{z\{1 - \Phi(z)\}}{2(\sqrt{2} - 1)} \leq \frac{e^{-1/2}}{4(\sqrt{2} - 1)} < \frac{37}{100}, \qquad z = \frac{L}{\sqrt{V}},$$

using $1 - \Phi(z) \leq e^{-z^2/2}/2$. Reflection covers the bottom strict threshold. Equal diameters ($h_1 = h_2$) were already shown in Section 2 to have ratio $< 37/100$, whereas a single active coordinate achieves $C_1 > 37/100$.

3.2 Stationarity equations for unequal interior diameters

Scale the larger diameter to 1 and write the smaller diameter as $t \in (0, 1)$, with upper atom probabilities $p, q \in (0, 1)$. Set

$$a = p(1 - p), \qquad b = q(1 - q), \qquad r_p = p^2 + (1 - p)^2, \qquad r_q = q^2 + (1 - q)^2,$$
$$V = a + t^2 b, \qquad R = ar_p + t^3 br_q.$$

The uncentered support is $\{0, t, 1, 1 + t\}$, and the three remaining interior closed thresholds have data

Uncentered threshold	F	Centered threshold y
0	$(1 - p)(1 - q)$	$-p - tq$
t	$1 - p$	$-p + t(1 - q)$
1	$1 - pq$	$1 - p - tq$.

With $z = y/\sqrt{V}$, $D = F - \Phi(z)$, $C = DV^{3/2}/R$, $\alpha = \phi(z)/\sqrt{V}$, $A = (3D + z\phi(z))/(2V)$, and $\lambda = D/R$, an interior maximum on a selected branch must satisfy $g_p = g_q = G = 0$, where

$$\begin{aligned} g_p &= F_p + \alpha + A(1-2p) - \lambda(1-2p)^3, \\ g_q &= F_q + \alpha t + At^2(1-2q) - \lambda t^3(1-2q)^3, \\ G &= \phi(z)RB + 3Dwta\sqrt{V}\,(r_p - tr_q), \end{aligned} \tag{3.1}$$

and (w, B) for the three thresholds $0, t, 1$ is respectively

$$(1-q,\ p[(1-p) - t(1-q)]), \qquad (q,\ -p[(1-p) + qt]), \qquad (1-q,\ (1-p)[p + t(1-q)]).$$

3.3 Compact variance bound and certified exclusion

In variance-one units, probability stationarity of coordinate i means that

$$Q_i(u) = G_i(y-u) - F(y) + \phi(y)u + A(u^2 - s_i) - C\{|u|^3 - b_i - 3M_i u\}$$

takes equal values at the two support points of X_i; since $\mathbb{E}Q_i(X_i) = 0$, both values vanish. Using $Cb_i \le D \le 1$, $|M_i| \le s_i \le 1$, $\phi(y) < 2/5$, and $|A| \le 13/8$ gives $C|u|^3 \le 4 + (2/5 + 3C)|u| + (13/8)u^2$, so $|u| < 3 + 8/C$. At $C \ge 37/100$, the larger normalized diameter $1/\sqrt{V}$ is less than $2(3 + 800/37) = 1822/37$, which implies

$$V > \frac{1369}{3319684} > \frac{1}{2500}, \qquad R \ge \frac{V^{3/2}}{\sqrt{2}} > \frac{V}{75}. \tag{3.2}$$

The certificate `certificates/structure/two_arbitrary_gaps-base.json` (together with its continuation and 37,421-leaf independent arithmetic replay; see `certificates/structure/README.md`) covers $(p, q, t) \in [0,1]^3$ for all three interior thresholds, certifying that every box is excluded by (3.2), by $C \le D\sqrt{a/r_p^2 + b/r_q^2} < 37/100$, or by a nonzero interval enclosure of g_p, g_q, or G. Thus no interior stationary point can achieve $C \ge 37/100$, leaving only the one-coordinate maximizer with ratio $C_1 > 37/100$. Finally, adjoining a second symmetric two-point coordinate of diameter $\varepsilon \downarrow 0$ shows that the supremum over pairs with both variances positive remains C_1 and is unattained.

4 Binomial phase and the optimal number of zero coordinates

In this section we analyze the m-summand i.i.d. binomial maxima B_m and their padded envelope $P_N := \max_{1 \le m \le N} B_m$. By Theorem 1.1, $P_N = C_N^{\mathrm{com}}$ is the exact N-summand common-diameter maximum.

Put $s = \sqrt{10}$ and define the algebraic constants

$$\begin{aligned} p &= \frac{4-s}{2}, \qquad q = 1-p, \qquad t = s-3, \qquad u = p - t, \qquad c_* = \frac{20+s}{72}, \\ Q_* &= \frac{\phi(0)(867 + 287s)}{25920}, \qquad \alpha = \frac{\phi(0)(60 + 19s)}{30}, \qquad b = \frac{1010 - 397s}{12960}, \\ \delta &= c_* - b = \frac{2590 + 577s}{12960}, \qquad \eta = \frac{Q_*}{\alpha} = \frac{2510 - 747s}{8640}, \qquad \omega = \frac{7773 - 2447s}{8640}. \end{aligned} \tag{4.1}$$

For a general Bernoulli parameter $v \in (0,1)$, write $a(v) = v(1-v)$, $r(v) = v^2 + (1-v)^2$, and $\tau(v) = 1 - 2v$; for $x \in \mathbb{R}$, write $\langle x \rangle = x - \mathrm{round}(x)$.

Theorem 4.1 (Asymptotic zero padding and record gaps in P_N)**.** *For every sufficiently large integer N, $P_N = \max_{1\le m\le N} B_m$ is attained at a unique active count m_N, and the number of zero coordinates $j_N = N - m_N$ belongs to $\{0,1,2,3,4\}$. As $N\to\infty$, the limiting frequencies of $j_N = 0,1,2,3,4$ exist and equal*

$$p-\omega, \qquad p-\omega, \qquad t, \qquad \omega, \qquad \omega,$$

respectively. Moreover, the gaps between successive record active counts m_N eventually belong to $\{2,3,5\}$, and each of the three gaps $2,3,5$ occurs infinitely often with positive asymptotic density.

4.1 Uniform Edgeworth–Euler–Maclaurin expansion at an integer threshold

Let $F_{m,v}$ be the distribution function of $\mathrm{Bin}(m,v)$. For an integer k and $v\in(0,1)$, set $c = mv-k$, $d = (ma(v))^{-1/2}$, and $D_{m,k}(v) = F_{m,v}(k) - \Phi(-cd)$.

Lemma 4.2. *For every compact interval $J\subset(0,1)$ and $M>0$, uniformly for $|c|\le M$ and $v=(k+c)/m\in J$,*

$$D_{m,k}(v) = \phi(0)\big[d(1/2+\tau/6) + d^3C(\tau,-c) + d^5T_5(\tau,-c)\big] + O(d^7), \tag{4.2}$$

and the same asymptotic expansion holds upon differentiating with respect to c at fixed m,k (and extends to any fixed order in d). Here

$$\begin{aligned} C(\tau,\beta) &= \frac{-1080(1+\tau)\beta^2 - (90\tau^2+1080\tau+630)\beta + \tau^3 - 45\tau^2 - 189\tau - 135}{4320},\\ T_5(\tau,\beta) &= \frac{3+5\tau}{48}\beta^4 + \frac{27+60\tau+25\tau^2}{288}\beta^3 + \frac{15+31\tau+25\tau^2+\tau^3}{192}\beta^2\\ &\quad + \frac{549+1320\tau+1030\tau^2+120\tau^3+5\tau^4}{23040}\beta\\ &\quad + \frac{189+927\tau+126\tau^2+234\tau^3+21\tau^4-25\tau^5}{193536}. \end{aligned}$$

Proof. By Fourier inversion on $[-\pi,\pi]$ with $|1-v+ve^{\mathrm{i}\xi}|^m = (1-4a(v)\sin^2(\xi/2))^{m/2} \le e^{-2z^2/\pi^2}$ ($z=\xi\sqrt{ma(v)}$), expanding $\log \mathbb{E}e^{\mathrm{i}\xi(\mathrm{Bin}(m,v)-mv)}$ on $|z|\le m^{\varepsilon}$ gives a uniform local Hermite expansion for each point mass $\mathbb{P}(\mathrm{Bin}(m,v)=j)$ and its c-derivatives (using $\partial_c(1-v+ve^{\mathrm{i}\xi})^m = (e^{\mathrm{i}\xi}-1)(1-v+ve^{\mathrm{i}\xi})^{m-1}$) to error $O(d^{12})$. Summing over $j\le k$ via Euler–Maclaurin summation (with lattice corrections $df/2 + d^2f'/12 - d^4f'''/720$ and Bernoulli cumulant ratios $\tau, 1-6a, \tau(1-12a), 1-30a+120a^2, \tau(1-60a+360a^2)$) leaves only odd powers of d at $-cd$ and yields (4.2) (compare Schulz [7, Remark 4.3, p. 25] for the cubic polynomial $C(\tau,\beta)$). □

Dividing (4.2) by $r(v)d$ gives the branch ratio expansion

$$R_{m,+,k}(v) = E(v) - \frac{Q(v,c)}{m} - \frac{H(v,c)}{m^2} + O(m^{-3}), \tag{4.3}$$

where

$$E(v) = \phi(0)\frac{3+\tau(v)}{3(1+\tau(v)^2)}, \qquad Q(v,c) = -\frac{\phi(0)C(\tau(v),-c)}{r(v)a(v)}, \qquad H(v,c) = -\frac{\phi(0)T_5(\tau(v),-c)}{r(v)a(v)^2}.$$

For each fixed v, the quadratic function $c\mapsto Q(v,c)$ attains its minimum at $c_0(v) = (\tau(v)^2 + 12\tau(v)+7)/[24(1+\tau(v))]$, with minimum value

$$Q_{\min}(v) = \frac{\phi(0)\{345+72\tau-498\tau^2-8\tau^3-23\tau^4\}}{4320(1+\tau)(1-\tau^4)}. \tag{4.4}$$

4.2 Second-order expansion of B_m and the five-state phase envelope

By Proposition 6.1 and pp. 29–31 of Schulz [7], noncentral thresholds and parameters $v \notin [1/6, 5/6]$ cannot compete with $C_{\mathrm{E}} - O(m^{-1})$. After reflecting to $v \le 1/2$, $E(v)$ has its unique maximum C_{E} at $v = p$ with $E(v) = C_{\mathrm{E}} - \alpha(v-p)^2 + O((v-p)^3)$, and the positive closed central threshold dominates the left-limit central threshold (whose leading term has $1/2 - \tau/6 < 1/2 + \tau/6$). Comparing (4.3) with the trial choice $v = (k + c_*)/m$ (for $k = \mathrm{round}(mp - c_*)$) gives $v - p = O(m^{-1})$ and $c - c_0(v) = O(m^{-1/2})$, and branch stationarity $\partial_c R_{m,+,k} = 0$ refines this to $c - c_0(v) = O(m^{-1})$ and $v = (k + c_*)/m + O(m^{-2})$. Setting $e = k + c_* - mp$ and completing the square in $\alpha e^2 + Q_p(p, c_*)e + H(p, c_*) = \alpha(e - b)^2 + H_*$ yields the uniform second-order expansion

$$B_m = C_{\mathrm{E}} - \frac{Q_*}{m} - \frac{H_* + \alpha\,\mathrm{dist}(mp - \delta, \mathbb{Z})^2}{m^2} + O(m^{-3}). \tag{4.5}$$

Because $B_m < C_{\mathrm{E}}$ for all m and $B_m \to C_{\mathrm{E}}$, any maximizing active count m_N for P_N satisfies $m_N \to \infty$, and comparing (4.5) at m_N and N forces $j_N = N - m_N = O(1)$. With $\theta_N = \mathrm{round}(Np - \delta) - (Np - \delta) \in [-1/2, 1/2]$, we have for bounded $j \ge 0$:

$$N^2(B_{N-j} - B_N) = \alpha\{\theta_N^2 - \mathrm{dist}(\theta_N + jp, \mathbb{Z})^2 - \eta j\} + O(N^{-1}).$$

Maximizing B_{N-j} therefore minimizes $f_j(\theta) = \eta j + \mathrm{dist}(\theta + jp, \mathbb{Z})^2$. Define the five transition points in $[-1/2, 1/2]$ by

$$\begin{aligned} T_{13} &= -\frac{p+u}{2} + \frac{\eta}{t}, & T_{30} &= -\frac{u}{2} - \frac{3\eta}{2u}, \\ T_{02} &= \frac{t}{2} + \frac{\eta}{t}, & T_{24} &= \frac{3t}{2} + \frac{\eta}{t}, \\ T_{41} &= \frac{q+2t}{2} - \frac{3\eta}{2u}. \end{aligned} \tag{4.6}$$

On $[-1/2, 1/2]$, the lower envelope $\min_{j \ge 0} f_j(\theta)$ is achieved on six intervals by $j \in \{0, 1, 2, 3, 4\}$:

Phase interval for θ_N	Optimal zero count j
$[-1/2, T_{13}]$	1
$[T_{13}, T_{30}]$	3
$[T_{30}, T_{02}]$	0
$[T_{02}, T_{24}]$	2
$[T_{24}, T_{41}]$	4
$[T_{41}, 1/2]$	1

Every $j \ge 5$ satisfies $f_j(\theta) \ge 5\eta$, whereas $\min_{0 \le j \le 2} f_j(\theta) \le 2\eta + p^2/4 < 5\eta - 59/10000$ on $[-1/2, 1/2]$. Hence $j_N \in \{0, 1, 2, 3, 4\}$ for all large N, and by Weyl equidistribution of the irrational rotation θ_N, the limiting frequencies of $j_N = 0, 1, 2, 3, 4$ equal the lengths $p - \omega, p - \omega, t, \omega, \omega$ of the corresponding phase intervals.

4.3 $O(N^{-1})$ boundary drift and quadratic-field separation

To resolve ties near the five transition points (4.6), carrying Lemma 4.2 to order d^7 gives the third-order expansion on a winning branch:

$$B_m = C_{\mathrm{E}} - \frac{Q_*}{m} - \frac{F_2(e)}{m^2} + \frac{G(e) - J_*}{m^3} + O(m^{-4}), \tag{4.7}$$

where $F_2(e) = \alpha e^2 + Q_p e + H_0$, $K(e) = (2\alpha + Q_{pc})e + Q_p + H_c$, and $G(e) = E_3 e^3 - \frac{1}{2}Q_{pp}e^2 - H_p e + K(e)^2/(2Q_{cc})$. Expanding B_{N-j} in powers of N^{-1} shows that the crossing between parabolas i and j occurs at $\theta = T + D_{ij}/N + O(N^{-2})$, where explicit evaluation gives

$$\begin{aligned} D_{13} = D_{02} = D_{24} = D_A &= \frac{129918319}{529079040} + \frac{191267395}{1693052928}s, \\ D_{30} = D_{41} = D_B &= -\frac{663979357}{2351462400} - \frac{20072514361}{141087744000}s. \end{aligned} \tag{4.8}$$

For each of the five transition points T, writing $h_N = \theta_N - T = (A_N + B_N s)/L_0$ with $L_0 = 25920$ and $A_N, B_N \in \mathbb{Z}$, the field norm $\nu_N = (A_N^2 - 10B_N^2)/L_0^2 \in L_0^{-2}\mathbb{Z}$ is bounded whenever $|h_N| \le C/N$, and $Nh_N = -\nu_N/s + O(N^{-1})$. Because both D_A and D_B have nonzero rational parts in $\mathbb{Q}(s)$, $-\nu_N/s$ is bounded away from D_A and D_B, so $|\theta_N - T - D_{ij}/N| \ge \varepsilon/N$ for all large N. This dominates the $O(N^{-2})$ remainder and proves that m_N is unique for all large N. Finally, the exact translation identities $T_{02} - p = T_{13}$, $T_{13} + q = T_{24}$, and $T_{30} + q = T_{41}$ preserve the drift classes D_A and D_B, so under $\theta_{N+1} = \langle \theta_N - p \rangle$ the transitions of j_N follow the directed graph

$$0 \longrightarrow 1, \qquad 1 \longrightarrow 0 \text{ or } 2, \qquad 2 \longrightarrow 0 \text{ or } 3, \qquad 3 \longrightarrow 4, \qquad 4 \longrightarrow 0,$$

whose return cycles to $j_N = 0$ have lengths $2, 3, 5$ with positive limiting frequencies $u - \omega, t - \omega, \omega$. This completes the proof of Theorem 4.1.

5 Local replacement of nearby support diameters

In this section we prove Theorem 1.3. Dividing all summands by the variance-weighted mean diameter h_0, we set

$$q_i = 1 - p_i, \qquad r_i = p_i^2 + q_i^2, \qquad R_0 = \sum_{i=1}^n a_i r_i, \qquad \mu = \sum_{i=1}^n p_i, \qquad \ell = \sum_{i=1}^n a_i |e_i| \le \sqrt{A}\,\sigma,$$

where $\sum_{i=1}^n a_i e_i = 0$ and $A/2 \le R_0 \le A$. Writing $K = \sum_{i=1}^n B_i$ and $T = \sum_{i=1}^n e_i(B_i - p_i)$, the centered sum is $S = K - \mu + T$ with variance $A + \sigma^2$. Set $c_k = \mathbb{P}(K = k)$ and $N_j(k) = \mathbb{E}[T^j \mathbf{1}_{\{K=k\}}]$.

5.1 Probability-free Poisson-binomial and weighted-moment bounds

Uniformly over all probabilities $p_i \in (0,1)$ and all $k \in \mathbb{Z}$, as $A \to \infty$,

$$\begin{aligned} c_k &= \frac{\phi(x_k)}{\sqrt{A}} + O(A^{-1}), \\ F_K(k) - \Phi(x_k) &= \frac{\phi(x_k)}{\sqrt{A}} \left\{ \frac{1}{2} + \frac{\gamma_3}{6}(1 - x_k^2) \right\} + O(A^{-1}), \\ x_k &= \frac{k - \mu}{\sqrt{A}}, \qquad \gamma_3 = \frac{1}{A}\sum_{i=1}^n a_i(1 - 2p_i) \in [-1, 1], \end{aligned} \tag{5.1}$$

and the unconditioned weighted moments $N_j(k) = \mathbb{E}[T^j \mathbf{1}_{\{K=k\}}]$ satisfy

$$\begin{aligned} |N_1(k)| &\le C\ell A^{-3/2}, \\ |N_2(k) - \sigma^2 c_k| &\le \frac{C\sigma^2}{A}, \\ 0 \le N_4(k) &\le \frac{C(\sigma^4 + \eta^2\sigma^2)}{\sqrt{A}}. \end{aligned} \tag{5.2}$$

To prove (5.2) without dividing by individual characteristic functions, write $f_i(t) = q_i + p_i e^{\mathrm{i}t}$, $s = e^{\mathrm{i}t} - 1$, $\Psi = \prod_{i=1}^n f_i$, and Ψ_i, Ψ_{ij} for the products with one or two factors omitted. Because $\sum_{i=1}^n a_i e_i = 0$, the Fourier transform of N_1 is $\widehat{N}_1(t) = s\sum_i a_i e_i \Psi_i = -s^2 \sum_i a_i p_i e_i \Psi_i$, whose $L^1([-\pi,\pi])$ norm is $O(\ell A^{-3/2})$. Similarly, using $\Psi_{ij} = \Psi + (1-f_i)\Psi_i + (1-f_j)\Psi_j + (1-f_i)(1-f_j)\Psi_{ij}$ and $u_i = a_i e_i$ ($\sum_i u_i = 0$) bounds $\|\widehat{N}_2 - \sigma^2\Psi\|_1 = O(\sigma^2/A)$, and expanding T^4 into index partitions $(4), (3,1), (2,2), (2,1,1), (1,1,1,1)$ bounds $\|\widehat{N}_4\|_1 = O((\sigma^4 + \eta^2\sigma^2)/\sqrt{A})$.

5.2 Threshold localization prior to conditioning

At any threshold $y = k - \mu + \theta$ with $k = \mathrm{round}(y + \mu)$ and $|\theta| \le 1/2$, decomposing $\{S \le y\}$ by the value of K and summing $N_4(j)/(|j-k| - 1/2)^4$ over $j \ne k$ gives

$$F_h(y) = F_K(k) - c_k \mathbb{P}(T > \theta \mid K = k) + E_{k,\theta}, \qquad |E_{k,\theta}| \le \frac{C(\sigma^4 + \eta^2\sigma^2)}{\sqrt{A}}, \tag{5.3}$$

while the third-moment sum R_h satisfies

$$R_h - R_0 = 3\sum_{i=1}^n a_i r_i e_i + 3\sum_{i=1}^n a_i r_i e_i^2 + \sum_{i=1}^n a_i r_i e_i^3, \qquad |R_h - R_0| \le C(\ell + \sigma^2 + \eta\sigma^2). \tag{5.4}$$

Thus any threshold achieving $\mathcal{C}_h(y) \ge c_0$ has bounded $|x_k|$, so $c_k \ge c_{c_0}/\sqrt{A}$, and dividing (5.2) by c_k yields

$$\begin{aligned} &|\mathbb{E}(T \mid K = k)| \le \frac{C\ell}{A}, \\ &\mathbb{E}(T^2 \mid K = k) = \sigma^2\{1 + O(A^{-1/2})\}, \\ &\mathbb{E}(T^4 \mid K = k) \le C(\sigma^4 + \eta^2\sigma^2). \end{aligned} \tag{5.5}$$

5.3 Conditional L^1 lower bound via rejective-sampling matchings

We next prove that at any localized count k,

$$\mathbb{E}(|T| \mid K = k) \ge \frac{c\ell}{\sqrt{A}}. \tag{5.6}$$

The conditional law of $(B_1, \dots, B_n)$ given $K = k$ is a rejective sampling law [3, Section 2]; by (5.1), for any set F of at most three indices, $\mathbb{P}(B_F = b_F \mid K = k) = \mathbb{P}(B_F = b_F)\{1 + O(A^{-1/2})\}$. Let B, B' be two independent draws from the conditional law given $K = k$, and let $C = \{i : B_i \ne B_i'\}$, which has even cardinality $m = |C|$. Conditional on C and on the common values outside C, the signs $B_i - B_i' \in \{-1, 1\}$ for $i \in C$ are uniformly distributed over balanced sign vectors (since the odds product $\prod_{i \in C}(p_i/q_i)$ is independent of which half of C is $+1$). Generating a uniform balanced sign vector by choosing a uniform perfect matching of C and assigning independent Rademacher signs to each matched pair gives, by Khintchine's inequality on each matching,

$$\mathbb{E}(|T - T'| \mid C) \ge \sqrt{\frac{2}{3}} \frac{\sum_{i<j \in C} |e_i - e_j|}{(m-1)\sqrt{m}} \qquad (m \ge 2).$$

Since $\mathbb{P}(i, j \in C) \ge c a_i a_j$ and $\mathbb{E}(m \mid i, j \in C) \le CA$, Jensen's inequality for $m^{-3/2}$ and $\sum_j a_j |e_i - e_j| \ge A|e_i|$ yield

$$2\mathbb{E}(|T| \mid K = k) \ge \mathbb{E}(|T - T'| \mid K = k) \ge cA^{-3/2} \sum_{i<j} a_i a_j |e_i - e_j| \ge \frac{c\ell}{\sqrt{A}},$$

proving (5.6).

5.4 Completion of the comparison and optimality of the rate

Setting $M_1 = \mathbb{E}(|T| \mid K = k)$, $W = \mathbb{P}(T > \theta \mid K = k)$, and $L = \theta + W$, equations (5.5)–(5.6) and Cauchy–Schwarz give $|\mathbb{E}(T \mid K = k)| \le M_1/4$, $M_1 \ge c\sigma^2/(\sigma+\eta)$, and

$$L \ge c \max\left(\frac{\ell}{\sqrt{A}}, \frac{\sigma^2}{\sigma+\eta}\right) > 0, \qquad |\theta| + \theta^2 + W \le CL, \qquad \sigma^2 \le C(\sigma+\eta)L. \tag{5.7}$$

Comparing $D_h(y) = F_h(y) - \Phi(y/\sqrt{A+\sigma^2})$ directly with the unperturbed discrepancy $D_0(k) = F_K(k) - \Phi(x_k)$ via (5.3) and (5.4) shows that every error term is $O((\sigma+\eta)^3 L + L/\sqrt{A})$, giving

$$\mathcal{C}_h(y) - \frac{A^{3/2}}{R_0} D_0(k) \le \{-c + C(\sigma+\eta)^3 + CA^{-1/2}\}L \le -\kappa \frac{\sigma^2}{\sigma+\eta},$$

which proves Theorem 1.3 (and when $p_i \in J \subset (0,1)$ for a compact set J, $\eta \le \sigma/\sqrt{\min_{p\in J} p(1-p)}$ gives $\sigma^2/(\sigma+\eta) \ge c_J \sigma$).

Finally, to see that the rate $\sigma^2/(\sigma+\eta)$ cannot be improved to σ without a lower bound on p_i, take a fixed large i.i.d. Esseen group of size m with ratio $> c_0$ attained on the positive closed branch, and adjoin one rare coordinate $B_{m+1} \sim \mathrm{Bern}(s)$ $(s \to 0)$ with diameter $g_s = 1 - \eta$ $(0 < \eta < 1/4)$ while giving the m group coordinates diameter $g_c = 1 + s(1-s)\eta/(ma)$. Because $0 < g_s/g_c < 1$, the closed CDF at uncentered threshold kg_c equals $(1-s)F_{\mathrm{group}}(k) + sF_{\mathrm{group}}(k-1)$, identical to the common-diameter value at $\eta = 0$. Since the smooth branch ratio is analytic in (s,η) near $(0,0)$ and is independent of η at $s = 0$, its mixed partial derivative is bounded, so $0 < \mathcal{C}_{\mathrm{com}}(s) - \mathcal{C}_h \le K_m s\eta$. Choosing $s = \eta^4$ gives $\sigma \sim \sqrt{s}\,\eta = \eta^3$ and gain $O(\eta^5) = O(\sigma^2/(\sigma+\eta))$.

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